COMPATIBILITY OF THE DERIVED PULLBACK WITH THE DERIVED TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the representatives are chosen explicitly and the comparison of the two paths is carried out with them, the general case following by the defining property of the representatives.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived pullback and its identification with the naive pullback on K-flat complexes with flat terms. Same chapter, the construction of Lf$\hat{}$* and the result that it agrees with f$\hat{}$* on K-flat complexes with flat terms. Supplies the replacements by which the two paths around the diagram are computed.
- Derived tensor product and the total complex of a double complex. Same chapter, the derived tensor product computed by K-flat representatives and the Tot construction. Supplies the four objects of the displayed diagram and the functoriality used to compare the two paths.

Source reference. Original lemma 06YS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3207–3207. Navigation pointer invoked at proof line 3791; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YY: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3585–3585. Navigation pointer invoked at proof line 3880; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0G7E: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3508–3508. Navigation pointer invoked at proof line 3793; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07A2: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3060–3060. Navigation pointer invoked at proof line 3909; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi. Let $\mathcal{K}^\bullet$ and $\mathcal{M}^\bullet$ be complexes of $\mathcal{O}_\mathcal{D}$-modules. The diagram*

$$\begin{array}{ccc} Lf^*(\mathcal{K}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_\mathcal{D}} \mathcal{M}^\bullet) & \longrightarrow & Lf^* Tot(\mathcal{K}^\bullet \otimes_{\mathcal{O}_\mathcal{D}} \mathcal{M}^\bullet) \\ \downarrow & & \downarrow \\ Lf^*\mathcal{K}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_\mathcal{C}} Lf^*\mathcal{M}^\bullet & & f^* Tot(\mathcal{K}^\bullet \otimes_{\mathcal{O}_\mathcal{D}} \mathcal{M}^\bullet) \\ \downarrow & & \downarrow \\ f^*\mathcal{K}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_\mathcal{C}} f^*\mathcal{M}^\bullet & \longrightarrow & Tot(f^*\mathcal{K}^\bullet \otimes_{\mathcal{O}_\mathcal{C}} f^*\mathcal{M}^\bullet) \end{array}$$

commutes.

Proof. We will use the existence of K-flat resolutions with flat terms (Lemma 06YS), we will use that derived pullback is computed by such complexes (Lemma 06YY), and that pullbacks preserve these properties (Lemma 0G7E). If we choose such resolutions $\mathcal{P}^\bullet \to \mathcal{K}^\bullet$ and $\mathcal{Q}^\bullet \to \mathcal{M}^\bullet$, then we see that

$$\begin{array}{ccc} Lf^*\mathrm{Tot}(\mathcal{P}^\bullet \otimes_{\mathcal{O}_\mathcal{D}} \mathcal{Q}^\bullet) & \longrightarrow & Lf^*\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}_\mathcal{D}} \mathcal{M}^\bullet) \\ \downarrow & & \downarrow \\ f^*\mathrm{Tot}(\mathcal{P}^\bullet \otimes_{\mathcal{O}_\mathcal{D}} \mathcal{Q}^\bullet) & \longrightarrow & f^*\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}_\mathcal{D}} \mathcal{M}^\bullet) \\ \downarrow & & \downarrow \\ \mathrm{Tot}(f^*\mathcal{P}^\bullet \otimes_{\mathcal{O}_\mathcal{C}} f^*\mathcal{Q}^\bullet) & \longrightarrow & \mathrm{Tot}(f^*\mathcal{K}^\bullet \otimes_{\mathcal{O}_\mathcal{C}} f^*\mathcal{M}^\bullet) \end{array}$$

commutes. However, now the left hand side of the diagram is the left hand side of the diagram by our choice of $\mathcal{P}^\bullet$ and $\mathcal{Q}^\bullet$ and Lemma 07A2. □